

# Moving Home: Non-Market Insurance and Labour Market Risk\*

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## Abstract

In 2023, more than half of renter households spent above 30% of their gross income on housing costs. This type of non-discretionary spending exacerbates consumption losses while unemployed. I use data from the Health and Retirement Study to show that workers use the option to move home as an insurance channel against labour market risk, and this channel is operative into middle age (up to age 40). To quantify this insurance mechanism, I estimate a structural lifecycle model of individuals who can insure against unemployment risk via saving or moving home. Agents exist in a directed search environment where they trade off future job market outcomes against job finding rates. I find that among workers who can move home, eliminating the option reduces average search duration by 8 weeks. Further, the move home option is equivalent in welfare terms to a 71% increase in the unemployment benefit for an ex ante consumer in an economy without co-residence.

**Keywords:** co-residence, shared housing, labour market risk, non-market insurance

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# 1 Introduction

Households of more than one adult generation have become common in the United States: in 2021, 18% of households had parents and adult children coresiding (Pew Research Center, 2022). One potential cause of this relationship is high housing costs: in 2023, more than half of renter households paid more than 30% of their gross income on housing (U.S. Census Bureau, 2024). An expensive housing market restricts disposable income and creates an environment where negative income shocks, such as unemployment, are more punishing for consumers. If an unemployed worker can avoid the cost of housing via a co-residence arrangement with parents, the welfare costs of unemployment fall and the choice to move home constitutes an insurance mechanism against labour market risk.

This paper documents a novel empirical trend in the share of adults entering co-residence with their parents and uses a quantitative model to estimate the welfare gain of the option to move home in the context of existing unemployment insurance. I find workers are approximately twice as likely to move home when they are moving from employment to unemployment relative to remaining employed. This trend is robust to both parent and child characteristics, and is observed into middle age for the children. Motivated by this finding, I estimate a structural model of job search and family co-residence to identify changes in search behaviour and welfare implications of the option to move home. Consumers with the option to co-reside search in submarkets that provide higher wages at the cost of longer search durations. For an ex ante consumer, the move home option is approximately equivalent in welfare terms to an approximately seventy percent increase in the UI benefit.

Existing work from Kaplan (2012) uses the National Longitudinal Survey of Youth (1997) to identify this insurance channel among young men. A major challenge in extending this understanding to older workers is disentangling the motives for young people moving home. Shared housing acts as an in-kind transfer from parents to their adult children by reducing housing costs and diminishing the impact of negative income shocks. However, co-residence can also represent an in-kind transfer from children to their parents if they provide household labour and especially eldercare that the parent would otherwise need to purchase on the market. The degree to which parental needs are a large concern and a confounding factor for identifying the labour market insurance mechanism is increasing in the ages of parents and children. Applying this insurance channel to a sample of workers into middle age requires controlling for the characteristics of aging

parents.

I leverage the Health and Retirement Study, a panel dataset of Americans over 50 and their linked children. The focus of the HRS is health and aging, making it an imperfect tool for analysis of earnings risk and unemployment. However, the HRS has several key features that are essential for addressing the question of shared housing as labour market risk. First, it explicitly links parents and adult children, including when those children live away from their parents. Many datasets, administrative and otherwise, link parent and child records while they share a dwelling, but those links weaken or break as children move out. Second, it maintains these linkages over a period of approximately 20 years with a consistent set of observables. A comparable dataset with parent-child linkages is the NLSY97, which tracks parents and children beginning in 1997 to the present. While these two products cover similar timeframes, many of the linked variables such as co-residence are gradually dropped from the NLSY97 as the sample cohort ages. This makes the NLSY97 an obvious candidate for evaluating the co-residence flows and labour market outcomes among young adults, but it rapidly loses its efficacy as the population of interest increases in age.

Third and crucially, the HRS reports increasingly important parent characteristics as they age including current diagnoses, health history, limitations to work and to living independently. Further, it reports on characteristics of the child-side of this relationship via the child's assistance with various eldercare needs and stated intentions to help in the future as their parents continue to age. More traditional support in the form of direct financial support is also present in both directions – from parents to children and vice versa. Information on each child's physical proximity to their parent is also included. With all of these characteristics combined, I can robustly estimate the effect of a child's Employed-to-Unemployed transition on the likelihood of them moving home with a parent.

I use these empirical observations to motivate a quantitative model of piece-rate job search and co-residence. In this environment, consumers vary in their individual-specific productivity and asset accumulation, which is tied to a savings choice. Working age individuals choose a submarket in which to search in every period. These submarkets are identified on three dimensions: the worker's age, which is observable by the firm and relevant for expectations of future match output; the worker's productivity; and the piece-rate of match output which constitutes the wage of an offer. Additionally, workers also choose whether to live at home with a parent or independently until age 40. After that time, the option to move home is withdrawn as parents begin to face mortality risk. When able, children trade off some utility benefit of living independently

against a financial cost of purchasing housing services on the private market. They additionally trade off wages and unemployment risk via their search choice; submarkets with higher job-finding probabilities offer lesser piece-rates and vice-versa.

Parameters estimated internally to the model target moments from the U.S. Bureau of Labor Statistics, U.S. Bureau of Economic Analysis, and earnings distribution estimates from Kuhn, Ríos-Rull, et al., 2016. One particularly important estimate in this environment is the utility benefit of living independently, for two reasons. First, this utility benefit is central to the main trade-off in terms of co-residence choice. Second, it is not directly observable or quantifiable in financial terms like the cost of housing. This benefit is calibrated to target the aggregate share of workers in the Health and Retirement Study who move home on a job separation. The model can then reproduce the age profile of this share who move home on an Employment-to-Unemployment transition.

When consumers have the option to move home, I find they select into submarkets with higher expected wages but a greater risk of a failed search and in turn, longer search durations. This is indicative of the labour market insurance mechanism influencing search behaviour and employment outcomes. Among workers who can move home, restricting that option reduces their average job search duration by approximately 8 weeks and is accompanied by an increase in the job finding and employment rates. While there are positive employment affects to such a counterfactual, equilibrium allocations for mean consumption and assets fall.

I evaluate the welfare implications of the move home option by comparing two model economies: one at the calibrated baseline and another in an environment where all workers must live independently and pay a fixed cost of housing. First, welfare from the move home option is largest when UI benefits are low. This indicates complementarity in the ability of workers to insure via UI benefits or co-residence. In the aggregate, the welfare benefits from the option to move home are large: in ex ante terms the option to move home is equivalent to a 71% increase in the UI benefit. However, this welfare value is also persistent along the lifecycle: workers aged 36-40 value the option equivalent to a 79% increase in the UI benefit. This result is particularly significant because in this literature most evidence pertains to young adults. However, by using a longer-term lifecycle model with co-residence into middle-age I can identify these benefits among older workers.

This paper contributes to a developing literature on co-residence between adult children and their parents. Empirically, evidence across multiple countries confirms the young adults use the option to move home with

parents as a form of insurance (Acolin et al., 2024; Albanesi et al., 2022; García-Andrés et al., 2021). However, the co-residence motives of older children are unexamined. Through the Health and Retirement Study, I am able to control for parent health needs and child eldercare, allowing identification of the labour market insurance motive later into the lifecycle, regardless of gender, education, or other child characteristics.

Existing work in quantitative macroeconomics tends to take one of two approaches when modeling the co-residence choice among adult children. The first approach is to construct a strategic game between parents and children, and characterize the effects of economic characteristics on the sequential choices of both agents. Models of this kind can be considered as evaluating behaviour under a joint coresidence choice in static and dynamic settings (Kaplan, 2012; Zaber, 2018). In contrast, the second approach is to abstract from the strategic interaction between parents and children to evaluate the existence of the coresidence option on some other macroeconomic phenomenon, usually in a dynamic setting (Dyrda et al., 2024). This paper takes the second approach, extending existing work by endogenizing labour supply decisions over the lifecycle in a directed search environment. Further, I estimate the value associated with the option to move home in the context of existing unemployment insurance programs.

The rest of this paper proceeds as follows: section 2 provides an overview of the Health and Retirement Study, identification of the labour market co-residence motive, and compares the main empirical result in the HRS to analogous trends in the NLSY97. Section 3 outlines the theoretical model of job search and non-market co-residence and section 4 explains the model estimation. Section 5 discusses the model validation and results; section 6 concludes.

## 2 Empirical Motivation

A primary challenge to identifying this housing-based insurance mechanism is isolating motives for migration into co-residence. One can imagine many reasons for adult children and parents to co-reside, and they are not limited to unemployment risk. From the perspective of the adult child, moving home can be the response to a divorce or separation, particularly when the outside option for housing is expensive. Considering aging parents, co-residence may arise as a response to growing complex health needs. To disentangle these factors from the labour market insurance motive I require data that reports these additional factors as well as links parents and children across a variety of ages.

## 2.1 Data

The Health and Retirement Study (HRS) is a panel of longitudinal survey data centered around aging adults and their families. It is maintained by the University of Michigan and the National Institute on Aging. The study surveys approximately 20,000 households every two years beginning in 1998. One relative strength of the HRS is repeated linkages between parents and children, particularly as the children age alongside their parents. Additionally, the survey reports a variety of non-standard parent characteristics such as health conditions, eldercare needs, and financial support to adult children. Using the health and eldercare information, I can isolate the motive among children to move home to be conditional on labour market outcomes exclusively. Identifying this insurance mechanism is the primary empirical contribution of this paper.

For convenience, I use the HRS Family File cleaned by the RAND Corporation which links the 20,000 parent households to 55,000 children up to 2018. This setting allows me to avoid the co-residence and employment impacts of the COVID-19 global pandemic. Additionally, I restrict the HRS sample to exclude post-secondary students as I assume they interact with the labour market and its risk distinctly from those seeking full-time employment. To further control health and eldercare motives for co-residence I restrict child ages from 20 to 40 years of age.

Given this analysis is based on flows around co-residence and employment, key observable characteristics are constructed from two subsequent survey waves. For a worker to have moved home, it is not sufficient for them to live with a parent in any given wave. Workers must first live independently, then in a subsequent wave be living with one or both parents. Similarly, to observe job separations I use a cross-wave variable for workers moving from employment to unemployment as well as employment to employment. The data on employment states is noisy and infrequent, so I cannot say who among the Employed-Unemployed group were separated voluntarily (i.e. quits) versus involuntarily (layoffs) or who among the Employed-Employed have stayed with a consistent employer and who have experienced a job separation between survey waves but were able to find employment again. Since these observables depend on two waves for identification, I drop the first wave (1998) from the analysis.

Previous work on unemployment risk often leverages higher-frequency datasets such as the National Longitudinal Survey of Youth (1997 Cohort). HRS interviews are two years apart, however adult children are questioned on their co-residence status well into middle-age. This contrasts with existing work from the

NLSY97, where parental co-residence is reported until 2009, when the oldest members of the cohort were 29. This longer series of co-residence information allows me identify that an insurance channel previously only examined for young men is present for all workers up to age 40.

## 2.2 Summary Statistics

Table 1 reports summary statistics for children in the HRS aged 20–40. Each panel covers a distinct area of child characteristics, and differences between the full sample of children, EU, and EE children are informative of the relative importance of child characteristics on employment transitions. Note that because of the simultaneous timing when these characteristics are observed (a common survey wave) it is unclear if child characteristics influence employment transitions or vice versa. These relationships are estimated more formally in section 2.3.

Table 1: Summary Statistics for Children Aged 20–40 in the HRS

|                                     | All Children | EU Type   | EE Type   |
|-------------------------------------|--------------|-----------|-----------|
| <b>Panel A: Demographics</b>        |              |           |           |
| Age                                 | 32.1         | 31.9      | 33.2      |
| Years of Education                  | 13.6         | 13.0      | 12.2      |
| <b>Panel B: Co-residence</b>        |              |           |           |
| Coresidence Share                   | .111         | .155      | .184      |
| Move Home Share                     | .018         | .057      | .020      |
| <b>Panel C: Financial Transfers</b> |              |           |           |
| Share who Give to Parents           | .018         | .020      | .015      |
| Share who Receive from Parents      | .215         | .233      | .163      |
| Mean Transfer from Parents          | \$1905.44    | \$1671.21 | \$1292.77 |
| <b>Panel D: Eldercare</b>           |              |           |           |
| Share who Help with ADLs            | .006         | .010      | .015      |
| Share who Help with IADLs           | .009         | .015      | .021      |
| Share who Intend to Help in Future  | .329         | .343      | .345      |

**Note:** Panel A reports mean child age and years of education. Panel B shows the static co-residence share and share of children who move home. Panel C lists the share of children who give and receive a transfer from their parent(s) as well as the size of the average transfer from a parent to their child. Panel D summarizes the share of children who assist with activities of daily living, instrumental activities of daily living, and intention to help with eldercare in the future.

In the first two panels of demographics and co-residence, there is little variation in most characteristics between EU, EE, and all children. However, the share of children who move home is much larger among EU children – suggestive of the key empirical observation summarized in Figure 1. Children transitioning

out of employment move home at a rate  $2.9\times$  larger than those who remain employed, before controlling for selection or further confounding characteristics.

One feature of the HRS that is useful when evaluating parental support is direct financial transfers between parents and children. This type of direct support also interacts with employment transitions since it functions as an insurance mechanism similarly to co-residence. EU children are 43% more likely to receive a transfer from a parent and on average receive around \$400 more than EE peers.

The final panel in Table 1 is informative of parent needs driving co-residence. It reports child assistance with activities of daily living and instrumental activities of daily living, as well as future intention to help. The former two are measures used by the National Institute of Health to summarize necessary capabilities for aging individuals to live independently. Activities of daily living refer to the most basic activities including feeding oneself, mobility within the household, and personal hygiene. Instrumental activities refer to more complex tasks such as transportation, managing finances, and meal preparation. Panel D shows there is little overlap between EU transitions and additional eldercare for parents: EE children are helping aging parents at higher rates than EU. The relative size of these shares are also small, peaking at around 2% of children aiding parents. This indicates that the age restriction on the sample of children filtering out most eldercare relationships while preserving co-residence as labour market insurance for children.

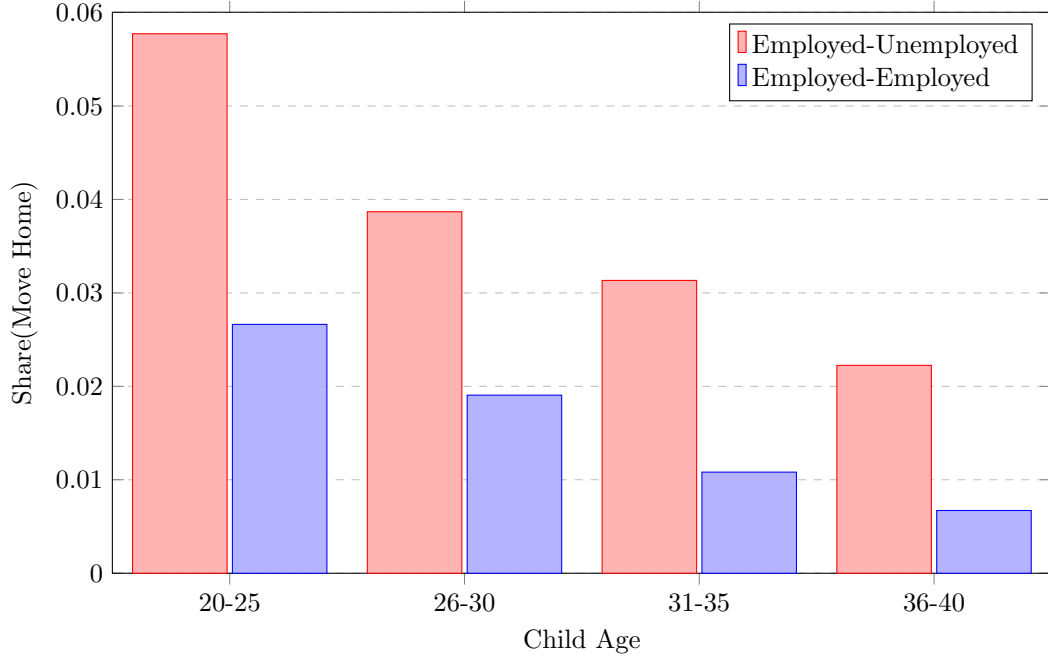
### 2.3 Labour Market Co-Residence Motive

The empirical contribution of this paper is centered around the observation that a larger share of adult children move home when experiencing a transition from employment to unemployment relative to peers who are persistently employed. This trend, shown in Figure 1, is representative of non-market insurance against labour market risk via in-kind housing services. Existing work from Kaplan (2012) identifies this mechanism for non-college men aged 17 to 22. In this section, I expand this understanding to all workers up to age 40 while controlling for parent and child characteristics.

To identify the existence of this labour market insurance channel via non-market housing arrangements, I employ a probit regression of cross-wave employment flows on workers' move home choice. A worker is considered to have "moved home" if in one survey period they are living independently and in a subsequent period are living with a parent. The primary covariates are employment flows: specifically for workers who are employed through two subsequent periods (EE) and those who separate from employment for any



Figure 1: Share who Move Home by Age and Employment Transition



**Note:** Adult children are more than twice as likely to move home when experiencing an EU transition relative to EE. Additionally, the share of EE and EU workers moving home is decreasing in age as children mature throughout the lifecycle. This is observed before controlling for parent or child characteristics.

reason (EU). Additional controls include child characteristics such as income, education, marital status, and financial transfers from parents as well parent characteristics including assets, home values, education, and health conditions.

After controlling for these characteristics, workers experiencing an EU transition are  $2.3\times$  more likely to move home with a parent relative to an identical worker with an EE transition. The likelihood of an adult child moving home is decreasing in their income and age, suggesting that workers in an E-U transition will self-insure via residual earnings from employment before they elect to move in with a parent.

A straightforward counterfactual can confirm the reverse case of the labour market insurance motive and is shown in Figure 2. Instead of moving home when experiencing a job separation, children in the data also move out when they find a job. Specifically, workers in an Unemployed-Employed transition move out of co-residence at much higher rates than those who remain unemployed.

Table 2: Probit Predicting Move Home Likelihood

| Independent Variable               | AME                    | p-value |
|------------------------------------|------------------------|---------|
| <b>Child Employment Transition</b> |                        |         |
| EE                                 | <i>Base</i>            |         |
| EU                                 | 0.016                  | 0.000   |
| UE                                 | 0.004                  | 0.040   |
| UU                                 | 0.003                  | 0.079   |
| <b>Child Income</b>                |                        |         |
| <10K                               | <i>Base</i>            |         |
| 10–35K                             | −0.012                 | 0.000   |
| 35–70K                             | −0.020                 | 0.000   |
| 70–100K                            | −0.020                 | 0.000   |
| 100K+                              | −0.024                 | 0.000   |
| <b>Child Age</b>                   |                        |         |
| 20–25                              | <i>Base</i>            |         |
| 26–30                              | −0.006                 | 0.001   |
| 31–35                              | −0.008                 | 0.000   |
| 36–40                              | −0.012                 | 0.000   |
| Parent Assets                      | $-1.84 \times 10^{-9}$ | 0.021   |
| Parent Home Value                  | $3.86 \times 10^{-9}$  | 0.002   |
| <b>Parent Employment</b>           |                        |         |
| Not Working                        | <i>Base</i>            |         |
| Part-time Work                     | 0.002                  | 0.457   |
| Full-time Work                     | 0.003                  | 0.008   |

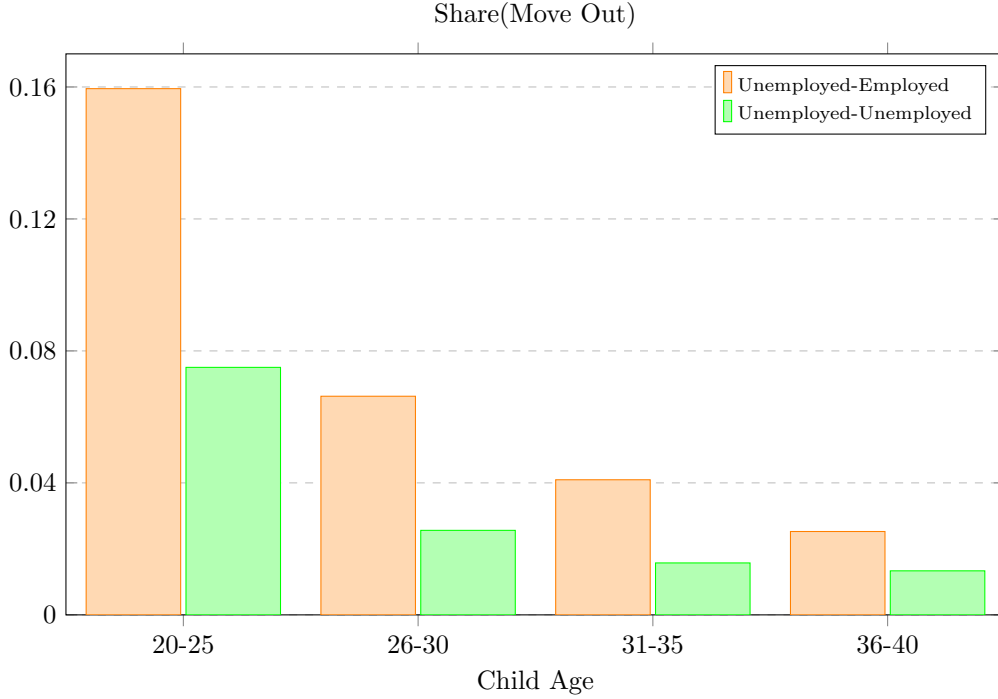
**Note:** Average marginal effects are reported as relative changes from the mean (0.0076). For the full regression specification and table of covariates, see appendix A.1.

## 2.4 Non-Labour Market Co-Residence Motives

Extensive sub-sampling is done on the raw HRS Family File to identify (1) the direction of those who move home, and (2) rule out non-labour market motives for moving home. I will begin with the former: ensuring that when adult children enter into co-residence they are in fact moving back in with their parents instead of hosting them in their own homes. This is related to the latter issue: the confounding nature of eldercare responsibilities on move home flows.

The first adjustment I make is to create a categorical variable for the inferred direction of co-residence based on home ownership. I assume that if, in the survey period when co-residence begins, the child is a homeowner and the parent is not, then the child is hosting the parent. In other words, co-residence

Figure 2: Move Out Flows Conditional on Employment Transitions



**Note:** Flows out of the parental household among job-finders (unemployed-employed) and those who remain persistently unemployed. For each age group, children transitioning into employment (UE) are more likely to move out. This demonstrates the reverse case of the labour market insurance motive.

happens because the parent has some need, be it financial, medical, or otherwise, instead of the child relying on parents as labour market insurance. In contrast, if the parent is a homeowner and the child is not, I assume the child is moving home in the sense they are returning to living with their parents after some period of independence. This scheme results in approximately two thirds of observations where a child enters co-residence having an identifiable direction; those left unidentified are excluded from the sample.

The question of eldercare is addressed in two ways. First, children are sampled on age as to exclude most parents with extensive eldercare needs. Details of this transformation are discussed in section 2.2. Second, I report average marginal effects from a probit estimation on the share of children who move home while controlling for eldercare and parent health characteristics in section 2.3.

## 2.5 NLSY97: Search Duration and Co-Residence

The NLSY97 reports the length of unemployment spells for youth moving between employers, which can be interpreted as a measure of search duration. Since employment and search are very noisy in the HRS, I leverage this feature of the NLSY97 to characterize the effects of co-residence on search behaviour. Intuitively, one can expect that children living at home are less financially constrained due to shared or avoided housing costs. This reduction of expenses affects the search process in two ways: first, workers may search less intensely, resulting in longer search times. Second, workers may reject otherwise viable offers if they are below their expected quality, both extending the search process but increasing wages conditional on a match.

Table 3: OLS Estimation of Co-Residence and Search Duration Among EU Workers

| Independent Variable     | Coefficient | p-value |
|--------------------------|-------------|---------|
| Co-Residence             | 1.389       | 0.034   |
| <b>Age</b>               |             |         |
| 20                       | 0.561       | 0.883   |
| 21                       | 1.584       | 0.677   |
| 22                       | 3.304       | 0.399   |
| 23                       | 3.203       | 0.425   |
| 24                       | 3.181       | 0.419   |
| 25                       | 3.294       | 0.409   |
| <b>Education</b>         |             |         |
| Less Than High School    | <i>Base</i> |         |
| HS Graduate              | −1.990      | 0.004   |
| Some College             | −1.548      | 0.446   |
| Post-Secondary and Above | −2.431      | 0.055   |
| Gender (Female)          | −0.970      | 0.055   |

**Note:** Living at home is associated with an additional 1.39 weeks of job search before finding a match. Coefficients on age reflect the relative change in search duration compared to a worker aged 18. For the full specification and results including year fixed-effects, see appendix A.2.

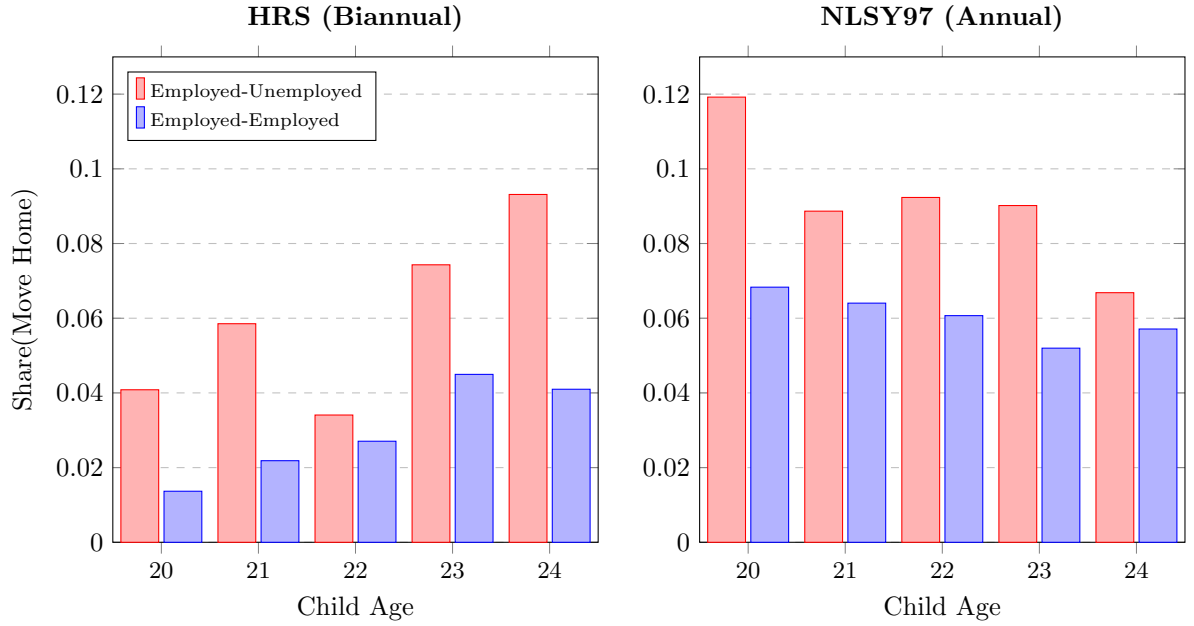
I estimate a simple ordinary least-squares regression of co-residence status on the average search duration in weeks. Coefficient estimates are shown in Table 3 and the most relevant is the marginal effect of co-residing with parents on search duration. After controlling for age, education, gender, and additional covariates, living at home is associated with an additional 1.4 weeks of search time before matching. The sample of workers is restricted to only those who have experienced an EU transition to account recent for job separation.

Education and sex are statistically significant in determining the duration of the search process, while age is not.

## 2.6 Robustness: HRS versus NLSY97

To evaluate the comparability of the two main datasets I use in this analysis, I present several key moments. Figure 3 plots the relative mass of workers moving home conditional on job separation (EU) or job retention (EE). In both datasets, EU workers are significantly more likely to move home than EE workers. This strengthens the argument for co-residence as labour market insurance, because it shows the main empirical observation in the HRS is also present in the more commonly used NLSY97.

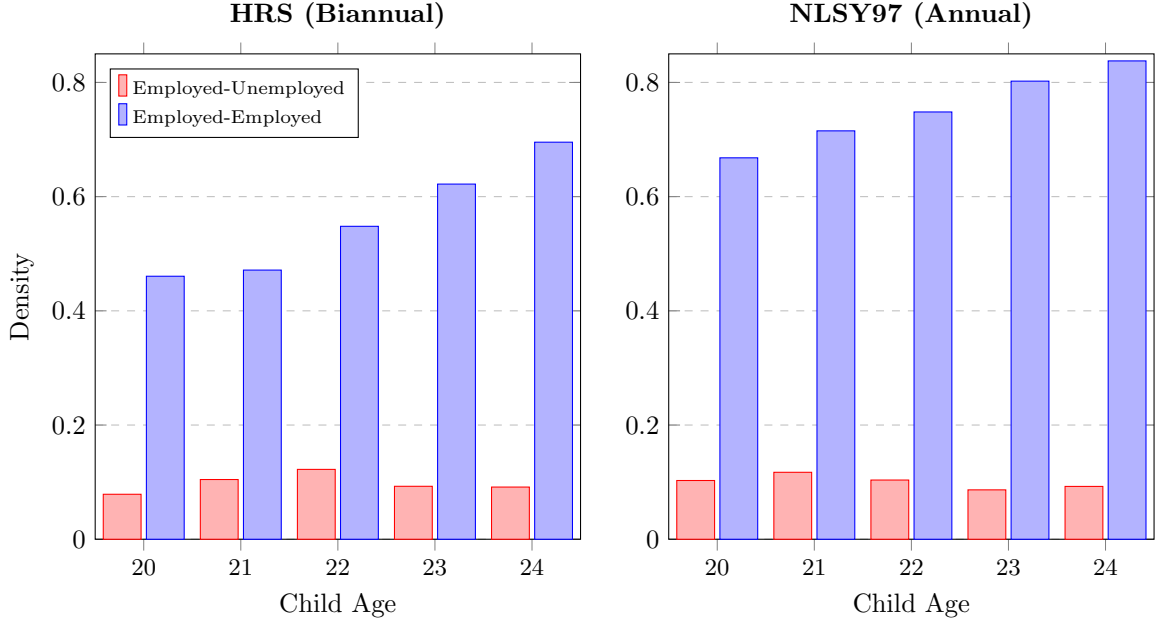
Figure 3: Comparison of Move Home Trends Among the HRS and NLSY97



**Note:** At each age from 20 to 24, workers experiencing a job separation (EU) are more likely to move home relative to their peers who remain employed (EE). Additionally, there is a decreasing trend in move home flows by age present in the NLSY97. This is also present in the HRS, but over a longer range of ages as shown in Figure 1.

Figure 4 plots the relative density of workers conditional on job transition. In both datasets, job tenure and retention increase as workers gain valuable experience. Similar to the previous plot in Figure 8, there

Figure 4: Comparison of Employment Transitions Between HRS and NLSY97 (2004)



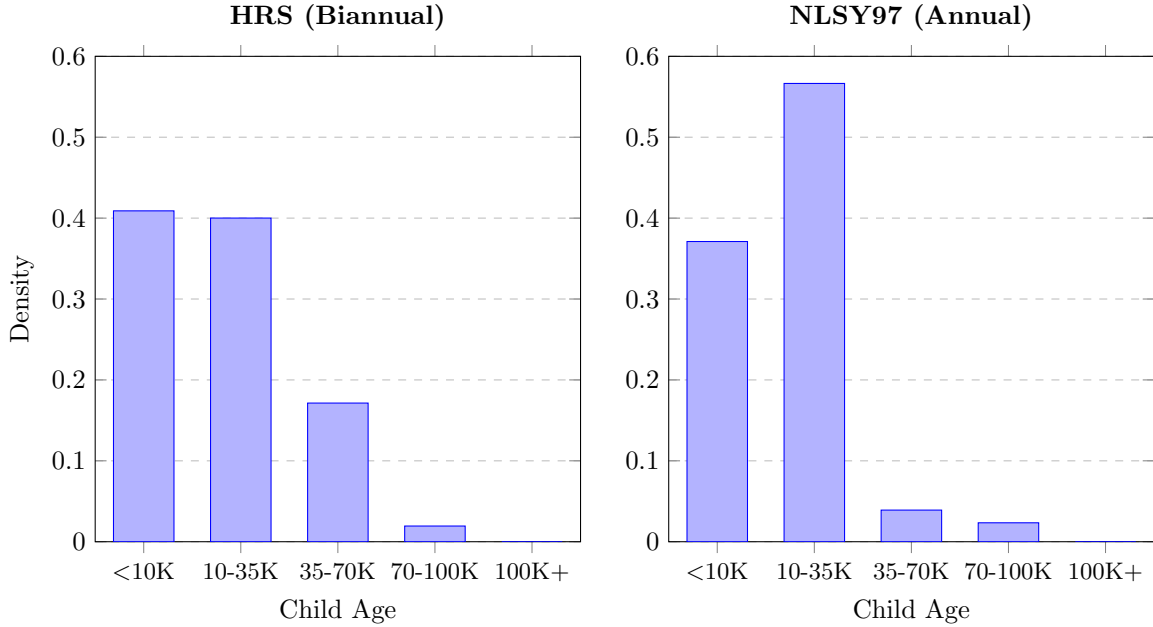
**Note:** As workers age, the relative mass of EE increases alongside job tenure. Estimates for EE densities vary significantly as a result of differences in survey composition and methodology. A comparison to a biannually-transformed NLSY97 is shown in appendix A.3.

is a distinct difference in levels between the two datasets. One possible explanation for this variation is differences in the frequency of the datasets. The HRS is limited to reporting once every two years, while the NLSY97 is primarily annual and in some cases monthly or weekly. Transforming the NLSY97 to match the frequency of the HRS reduces these differences in levels to some degree but inconsistencies remain.

Another explanation is variation in the sample composition between the two surveys. For example, the HRS sample is targeted to be representative of Americans over 50, while the NLSY97 is a representative sample of individuals born between 1980 and 1984. These surveys target very different populations of interest so it is unsurprising that some moments, notably employment flows, are very different. It is encouraging then that even with these differences in sample composition that two facts remain: first, there is suggestive evidence that workers use the option to move home as labour market insurance. Second, employment flows, at least with respect to increasing job tenure across the lifecycle, are consistent across both the HRS and

NLSY97.

Figure 5: Comparison of Income Among 20-25 Year-Olds in the HRS and NLSY97 (2004)



**Note:** Income distribution between both the HRS and NLSY97. The HRS reports child income as a categorical variable and I have converted the NLSY's continuous income to align the two datasets.

Income distributions from the two datasets share similar features. Figure 5 shows the income distribution among 20-25 year-olds in the year 2004. Incomes are binned to align with the categorization present in the HRS. In both datasets, very few workers report incomes above \$70,000 and virtually none earn above \$100,000. Respondents in the HRS are slightly richer, reflected in a larger density in the middle (\$35 – \$70K) bin relative to the second-to-lowest (\$10 – \$35K) category. Individuals with the lowest income are slightly more common in the HRS, but this difference is sufficiently small as to not be economically meaningful.

### 3 Model

This paper leverages a lifecycle model of consumption, savings, and co-residence. The main novel feature is the option workers have to move home, dependent on age. Additionally, consumers choose a submarket in which to search and trade-off against unemployment risk.

#### 3.1 Consumers

Consumers are the primary decision-makers in this model. They choose allocations for consumption ( $c$ ) and savings ( $a'$ ) at every age. During their working life and while unemployed, they choose a submarket in which to search indexed by piece-rate ( $\phi$ ) and their individual-specific productivity ( $\epsilon$ ). If matched, they earn an after-tax piece-rate of output as their wage; if unmatched, they receive a government-financed unemployment benefit  $b$ . When they retire, they receive a similarly funded pension benefit  $S$ .

All consumers in this environment are considered to be adult children in the context of multigenerational co-residence. This means that for each consumer there is some unmodelled agent, a parent, who is older than the child and provides non-market housing while they are able. When choosing where to live, consumers face a trade-off between independence utility ( $\chi$ ) and the cost of rental housing ( $\kappa_h$ ). If they live at home ( $d_h = 0$ ), they can avoid the cost of housing but forgo the utility benefits of living independently ( $d_h = 1$ ). This decision is subject to a Type-I extreme value taste shock ( $\xi^c, \xi^i$ ).

$$U(a, \epsilon, \phi, a', d_h) = \frac{c^{1-\sigma}}{1-\sigma} + d_h \chi \quad (1)$$

subject to

$$c + a' = (1 - \tau)\phi\epsilon + (1 + r)a - d_h \kappa_h$$

when employed, and

$$c + a' = b + (1 + r)a - d_h \kappa_h$$

when unemployed.

The problem for a worker can then be split into three cases: employed, unemployed, and searching for



work. Workers without a match search first, followed by choices conditional on their employment state. When searching, workers choose piece-rate which along with their age and individual-specific productivity determines a submarket  $(j, \epsilon, \phi)$ . The value of search is a comprised of the values of employment and unemployment, weighted by the job finding probability  $f(j, \epsilon, \phi)$ .

$$V_s(j, a, \epsilon) = \max_{\phi} \{f(\theta(j, a, \epsilon))V_u(j, a, \epsilon) + [1 - f(\theta(j, \epsilon, \phi))]V_e(j, a, \epsilon, \phi)\} \quad (2)$$

Whether employed or unemployed, the consumer chooses savings and whether or not to live at home until age 40. After age 40, they live independently<sup>1</sup> with certainty. Those with the option to move out face extreme value Type-I shocks on each option, co-residence and living independently, drawn from a common distribution. The value of unemployment prior to the realization of the shock is:

$$V_u(j, a, \epsilon) = \max \mathbf{E}_{\xi^c, \xi^i} \{V_u^{cores}(j, a, \epsilon) + \xi^c, V_u^{indep}(j, a, \epsilon) + \xi^i\} \quad (3)$$

The probability of a consumer choosing to live independently ( $h_u$ ) will depend on the relative values of  $V_u^{cores}$  and  $V_u^{indep}$  as well as the EV scale parameter  $\eta$  which determines the variance of the shock process.

$$Pr(d_h = 1|u) = h_u(j, a, \epsilon) = \frac{\exp(\eta V_u^{indep}(j, a, \epsilon))}{\exp(\eta V_u^{indep}(j, a, \epsilon)) + \exp(\eta V_u^{cores}(j, a, \epsilon))} \quad (4)$$

Following the realization of the shocks, the unemployed consumer chooses saving for the current period before re-entering the search process next period. This choice is summarized by a value function for each housing choice: co-residence and living independently. For an unemployed consumer who lives at home:

$$V_u^{cores}(j, a, \epsilon) = \max_{a' \geq 0} \left\{ \frac{c^{1-\sigma}}{1-\sigma} + \beta V_s(j+1, a', \epsilon) \right\} \quad (5)$$

subject to

$$c + a' = b + (1+r)a$$

When the unemployed worker lives independently, their problem is summarized as:

$$V_u^{indep}(j, a, \epsilon) = \max_{a' \geq 0} \left\{ \frac{c^{1-\sigma}}{1-\sigma} + \chi + \beta V_s(j+1, a', \epsilon) \right\} \quad (6)$$

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<sup>1</sup>This restriction allows parents to face mortality risk without creating “orphaned” children

subject to

$$c + a' = b + (1 + r)a - \kappa_h$$

The problem for an employed consumer is defined similarly. The consumer makes a discrete choice between co-residence and living on their own subject to extreme value shocks and their own age. The probability of such a consumer living independently is:

$$Pr(d_h = 1|e) = h_e(j, a, \epsilon, \phi) = \frac{\exp(\eta V_e^{indep}(j, a, \epsilon, \phi))}{\exp(\eta V_e^{indep}(j, a, \epsilon, \phi)) + \exp(\eta V_e^{cores}(j, a, \epsilon, \phi))} \quad (7)$$

where the discrete choice is given by:

$$V_e(j, a, \epsilon, \phi) = \max \mathbf{E}_{\xi^c, \xi^i} \{V_e^{cores}(j, a, \epsilon, \phi) + \xi^c, V_e^{indep}(j, a, \epsilon, \phi) + \xi^i\} \quad (8)$$

After observing the co-residence shocks, the employed worker chooses their savings before facing a job destruction shock  $\delta$ . If they are subject to the shock, they become unmatched and must search again in the following period. If they instead stay matched, they enter into employment next period without engaging in the search process. When an employed worker lives at home their value is:

$$V_e^{cores}(j, a, \epsilon, \phi) = \max_{a' \geq 0} \left\{ \frac{c^{1-\sigma}}{1-\sigma} + \beta[(1-\delta)V_e(j+1, a', \epsilon, \phi) + \delta V_u(j+1, a', \epsilon)] \right\} \quad (9)$$

subject to

$$c + a' = (1 - \tau)\epsilon\phi + (1 + r)a$$

As before, when living independently the employed worker's problem becomes:

$$V_e^{indep}(j, a, \epsilon, \phi) = \max_{a' \geq 0} \left\{ \frac{c^{1-\sigma}}{1-\sigma} + \chi + \beta[(1-\delta)V_e(j+1, a', \epsilon, \phi) + \delta V_u(j+1, a', \epsilon)] \right\} \quad (10)$$

subject to

$$c + a' = (1 - \tau)\epsilon\phi + (1 + r)a - \kappa_h$$

When consumers reach age 65, they enter retirement and receive a constant pension benefit  $S$ . Each period they make a savings decision and face an age-dependent mortality risk  $\psi_j$ . The problem of retirees

can be expressed as:

$$V_r(j, a) = \max_{a' \geq 0} \left\{ \frac{[S + (1+r)a - a']1 - \sigma}{1 - \sigma} + \beta \psi_j V_r(j+1, a') \right\} \quad (11)$$

### 3.2 Firms

A representative firm chooses whether or not to post in each submarket at a common cost  $\kappa_p$ , indexed by a required productivity for workers and a piece-rate of output that constitutes the wage. Additionally, the firm takes the worker's age into account when making the posting decision as it influences expectations of future output flows. Once a match is formed, it persists until a job destruction shock is received with probability delta or the worker reaches retirement age. All else equal, younger workers are more desirable to firms because they can produce output for longer before they retire. The firm's value can be expressed as:

$$V_f(j, \epsilon, \phi) = (1 - \tau)\epsilon(1 - \phi) + \beta[(1 - \delta)V_f(j+1, \epsilon, \phi) + \delta V_p(j+1, \epsilon, \phi)] \quad (12)$$

Where  $q(\theta(j, \epsilon, \phi))$  is the firm's job filling probability determined by the search process. Matching in this environment follows the tradition set by Den Haan et al. (2000), where the matching function is defined by:

$$M(u, v) = \frac{uv}{(u^\alpha + v^\alpha)^{1/\alpha}}, \quad \theta = \frac{v}{u}$$

where job filling and finding probabilities are:

$$q(\theta(j, \epsilon, \phi)) = \frac{M(u, v)}{v} = \left[ \frac{1}{1 + \theta^\alpha} \right]^{1/\alpha}$$

$$f(\theta(j, \epsilon, \phi)) = \frac{M(u, v)}{u} = \left[ \frac{1}{1 + (1/\theta)^\alpha} \right]^{1/\alpha}$$

A submarket will have vacancies posted as long as there are non-negative expected profits from operating. In other words, the firm must produce enough to cover the cost of posting. This free entry condition can be expressed as:

$$V_p(j, \epsilon, \phi) = \max\{q(\theta(j, \epsilon, a))V_f(j, \epsilon, \phi) - \kappa_p, 0\} \quad (13)$$

Given the production market is competitive, profits in equilibrium will be zero and  $V_p(j, \epsilon, \phi) = 0 \forall (j, \epsilon, \phi)$  submarkets where the firm posts.

### 3.3 Equilibrium

**Definition 1.** Given state vectors  $s_u = (j, a, \epsilon)$ ,  $s_e = (j, a, \epsilon, \phi)$ ,  $s_r = (j, a)$  and an exogenous interest rate  $r$ , an equilibrium in this model is characterized by policy functions  $c_u(s_u), c_e(s_e), c_r(s_r), h_u(s_u), h_e(s_e), a'_u(s_u), a'_e(s_e), a'_r(s_r), \phi(s_u)$  and stationary distributions of consumers  $\omega_u(s_u), \omega_e(s_e), \omega_r(s_r)$  such that:

1. The policy function  $\phi(s_u)$  solves the problem of the searcher given in (2)
2. The policy functions  $c_u(s_u), a'_u(s_u), h_u(s_u)$  solve the problem of the unemployed consumer given in (3)
3. The policy functions  $c_e(s_e), a'_e(s_e), h_e(s_e)$  solve the problem of the employed consumer given in (8)
4. The policy functions  $c_r(s_r), a'_r(s_r)$  solve the problem of the retired consumer given in (11)
5. A firm posts in each submarket where there is positive profit, subject to the free entry condition (13)
6. The government sets a proportional tax on labour income ( $\tau$ ) which satisfies their budget constraint

$$\int \tau \epsilon \phi d\omega_e(j, a, \epsilon, \phi) = \int b d\omega_u(j, a, \epsilon) + \int S d\omega_r(j, a) \quad (14)$$

## 4 Calibration

The two main externally estimated parameters for this model are the match elasticity ( $\alpha$ ), taken from the seminal paper by Den Haan et al. (2000) and the job destruction rate ( $\delta$ ) taken from the HRS as an annualized estimate for EU flows. Consumer risk aversion ( $\sigma$ ), discount factor ( $\beta$ ), and interest rate ( $r$ ) are taken at values standard from the literature. Series for survival probabilities are taken from the Social Security Administration's Bell and Miller (2005) life tables and the age profile of earnings is estimated in Raveendranathan and Stefanidis (2025) using the Panel Study of Income Dynamics.

A summary of the internal parameter estimates can be found in Table 4. The most important estimate from the perspective of the co-residence-based insurance is the independence utility,  $\chi$ , which is targeted to match the share of E-U workers who move home in the HRS. The labour market in this model is identified by

Table 4: External Parameters

| Parameter                     |                         | Value | Source                         |
|-------------------------------|-------------------------|-------|--------------------------------|
| <b>Panel A: Housing</b>       |                         |       |                                |
| $\kappa_h$                    | Rent to income ratio    | 0.05  |                                |
| $\eta$                        | EV distribution scale   | 1.000 |                                |
| <b>Panel B: Labour Market</b> |                         |       |                                |
| $\alpha$                      | Match elasticity        | 1.27  | Den Haan et al. (2000)         |
| $\delta$                      | Job destruction rate    | 0.02  | EU share (HRS)                 |
| $\nu$                         | Age profile of earnings |       | Panel Study of Income Dynamics |
| <b>Panel C: Lifecycle</b>     |                         |       |                                |
| $\sigma$                      | Risk aversion           | 2.00  |                                |
| $\beta$                       | Discount factor         | 0.96  |                                |
| $r$                           | Interest rate           | 0.04  |                                |
| $\psi$                        | Survival probabilities  |       | Social Security Administration |

the cost of posting,  $\kappa_p$ , targeted to the unemployment share and the standard deviation of productivity,  $\sigma_\epsilon$ , targeted to the standard deviation of log earnings for workers aged 26 to 30. The fiscal system is estimated via unemployment insurance ( $b$ ) and pension benefits ( $S$ ), which are targeted to aggregate spending on UI and pensions, respectively.

Table 5: Internal Parameters

| Parameter         |                         | Value | Target                                   | Model | Data  |
|-------------------|-------------------------|-------|------------------------------------------|-------|-------|
| $\chi$            | Independence utility    | 4.703 | EU move home share                       | .0413 | .0413 |
| $\kappa_p$        | Cost of posting         | .0034 | Unemployment share <sup>a</sup>          | .0478 | .0420 |
| $b$               | UI benefit              | .2454 | UI exp. to income ratio <sup>b</sup>     | .0042 | .0042 |
| $S$               | Social security benefit | .7537 | SS exp. to income ratio <sup>c</sup>     | .0525 | .0525 |
| $\sigma_\epsilon$ | St. dev. productivity   | .8868 | SD log earnings (age 26-30) <sup>d</sup> | .9000 | .9000 |

<sup>a</sup> U.S. Bureau of Labor Statistics (2025)

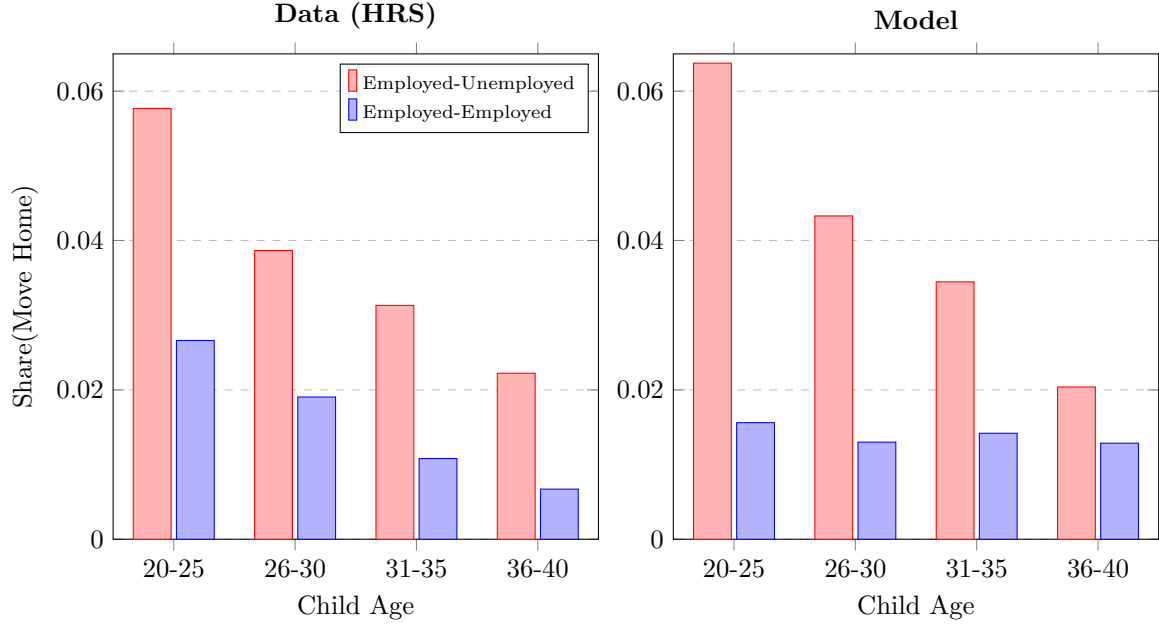
<sup>b,c</sup> U.S. Bureau of Economic Analysis (2024c)

<sup>d</sup> Kuhn, Ríos-Rull, et al. (2016)

## 5 Results

This section evaluates the quantitative model’s performance against the empirical trend of EU workers moving home in greater proportions than EE. Further, it outlines positive and normative results concerning job search behaviour, model aggregates, welfare in terms of unemployment insurance benefits, and heterogeneity

Figure 6: Comparison of Move Home Flows Between Model and Data



**Note:** Move home flows are larger for EU consumers and are decreasing in age.

in welfare by age.

## 5.1 Model Validation

The quantitative model is able to match two key features observed in the Health and Retirement Study, shown in Figure 6. First, workers experiencing an EU transition are significantly more likely to move home than peers in an EE transition. This is indicative of moving home acting as an insurance mechanism against these labour market transitions and associated earnings risk.

Second, the magnitude of this effect is decreasing in the worker's age, at least for the EU type. This has several implications. As workers age, they are more able to self-insure in their household through saving rather than co-residing. If living independently is sufficiently rewarding, EU workers will dissave before they move home. Another reason age interacts with the move home choice among workers is because it interacts with incomes via individual-specific productivity. As workers age, they become richer as their productivity

risks, but relatively poorer as they approach retirement and have fewer opportunities to match with a firm and earn income. Based on the declining nature of move home flows by age, it appears as though the increasing productivity effect outweighs the decreasing work duration effect.

## 5.2 Job Search Behaviour and Model Aggregates

Consumers who have access to the option to live at home search in submarkets with higher piece-rates and lower job finding probabilities. This is because unemployment is much less damaging in terms of individual utility when the cost of housing is avoidable by moving home with a parent. In this environment the worker can adjust their risk of unemployment by choosing a submarket in which to search.

Table 6: Model Aggregates with and without the Move Home Option

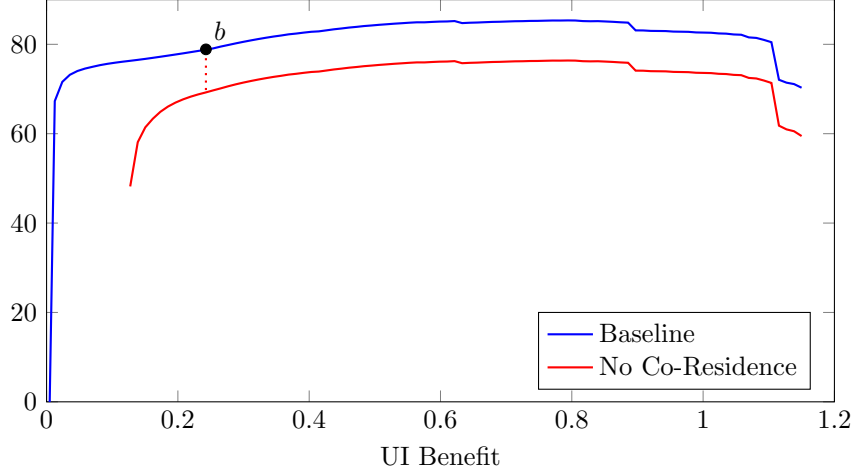
|                          | Baseline | No Co-Residence | Relative $\Delta$ (%) |
|--------------------------|----------|-----------------|-----------------------|
| Employment Share         | 0.95322  | 0.95623         | +0.316                |
| Average Job Finding Rate | 0.36288  | 0.38735         | +6.743                |
| Tax Rate                 | 0.06525  | 0.06492         | -0.505                |
| Average Assets           | 3.60543  | 3.60159         | -0.107                |
| Average Consumption      | 2.03555  | 2.03407         | -0.073                |

**Note:** Computed among workers aged 20-40 to control for the ability to move home. Restricting the move home option results in higher employment and job finding rates as well as lower taxes, assets, and consumption.

Table 6 details the impacts of individual-level choices on the full sample of workers aged 20 to 40. Constructing an environment where everyone must purchase housing increases employment through the job finding rate. This additional employment however does not lead to significantly increased asset accumulation or consumption – suggesting any financial gains from employment are absorbed by the mandatory cost of housing.

The most notable result is the over 2 percentage point increase in the job finding rate. The submarket choice, alongside savings, is the main mechanism consumers can control to influence their risk of and consumption loss from unemployment. The increase in the job finding rate in the absence of co-residence is equivalent to .1525 model periods (years) or approximately 8 additional weeks of search before a match.

Figure 7: Lifetime Expected Utility Conditional on UI Benefit



**Note:** The welfare associated with the move home benefit is largest when UI benefits are small. Note that no co-residence case is not well-defined for  $b \leq \kappa_h$ .

### 5.3 Welfare from Co-Residence

The vertical distance between the two plotted curves in Figure 7 is an estimate of welfare from the co-residence option. To evaluate this welfare in the context of an existing UI benefit, I measure the vertical distance ( $\sim 9$  utils) in terms of how much of a reduction in UI would induce the same utility reduction in an environment without co-residence. I find that ex ante the welfare from the option to move home is equivalent to a 71% increase in current UI benefits.

The relative welfare benefit of the move home option is largest at low levels of UI. For example, the vertical distance between the two curves is much larger below  $b$ , the calibrated UI benefit, than above. This relationship implies that unemployment insurance and the move home option are complementary insurance mechanisms against labour market risk. In an environment with a generous UI system, non-market insurance through shared housing is relatively less important for the welfare of workers. In contrast, when unemployment is more punishing through reduced benefits the welfare implications of moving home are significant.



## 5.4 Welfare Across the Lifecycle

In Section 2.3, I showed that co-residence as labour market insurance is operative for workers into middle age. In addition to this finding, the welfare benefits are also significant for older workers. Table 7 reports that the value of the move home option is large among all age groups, notably the oldest workers aged 36-40. This is additionally interesting when paired with the results from Figure 6, which shows that move home flows are declining in age in both model and data. It would appear as if older workers value the option to co-reside highly, but do not exercise it as often as younger workers. One reason that the move home option is exercised relatively less as workers age is because they become better able to insure without co-residence.

Table 7: Welfare by Age in Terms of UI Benefits

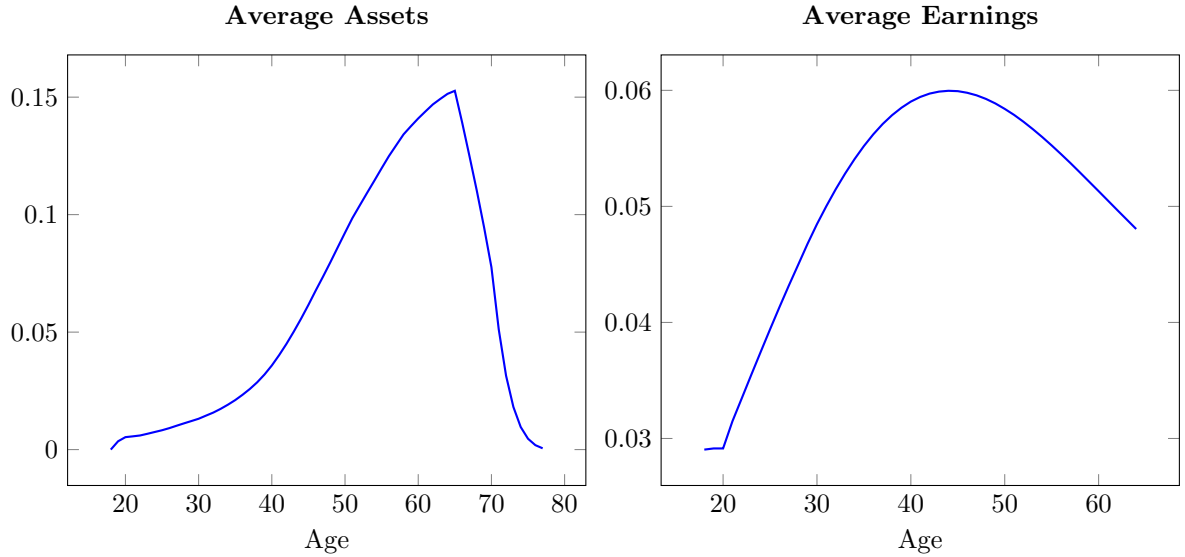
| Age   | Equivalent UI Benefit | Relative $\Delta$ (%) |
|-------|-----------------------|-----------------------|
| 20-25 | 0.13383               | 83.76                 |
| 26-30 | 0.12741               | 93.02                 |
| 31-35 | 0.12613               | 95.00                 |
| 36-40 | 0.13768               | 78.62                 |

**Note:** Estimates for welfare converted into an implied UI benefit. Column 2 presents the UI benefit that would make each age group indifferent between having the move home option versus being unable to move home in an economy with the baseline UI benefit (.245921). Column 3 shows the relative increase in the UI benefit that would provide equivalent welfare to allowing adult children to move home with their parents.

As workers age, they are on average better off in the current period. This is for two reasons: first, as shown in the first panel of Figure 8 they accumulate assets each period in which they work, and only begin to dissave when entering retirement. Second, the model includes an age component of individual-specific productivity to account for on-the-job experience. This age component of labour income can be seen in the second panel of Figure 8, where earnings are increasing into middle age but declining as retirement approaches. Falling wages later in life, however, are less damaging in welfare terms relative to early in life due to higher average assets.

This is not necessarily true when accounting for future expectations of workers. Since this model environment features a finite horizon lifecycle, the present expected utility of an individual is declining as they age and approach mortality. Additionally, in the absence of savings retirement will result in a large consumption loss relative to working periods. The potential negative effects of retirement on consumption create an environment where aging is ambiguous in welfare terms. Older agents can save more and have

Figure 8: Assets and Earnings by Age



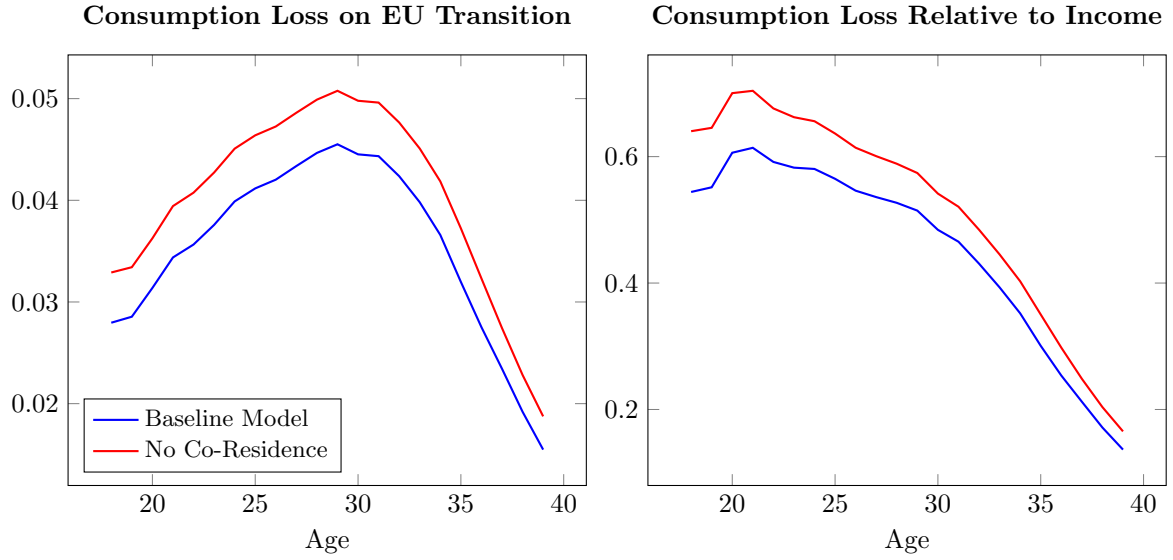
**Note:** Age profile of assets and earnings at the mean. Assets accumulate throughout the worker's life until they choose to dissave rapidly in retirement. Earnings are computed as after-tax labour income among the employed. Earnings rise early in life but decline as workers approach retirement for two reasons. First, the age component of productivity is sharply increasing at young ages before leveling off in late-middle age and declining slightly before retirement. Second, wages are partially determined via a posting decision by firms. Since age is both observable and relevant for future earnings flows, firms will naturally offer higher wages (piece-rates) to younger workers in expectation of output in future periods in the absence of a job destruction shock.

higher wages, however they have fewer additional years of work where they can accumulate assets and enjoy those heightened wages.

This explanation of the age dynamics of worker welfare is important for understanding Figure 9, which shows the consumption loss associated with an EU transition. When controlling for differences in income (Panel 2), consumption losses on unemployment are lowest among older workers, who in turn have the highest assets among those aged 20-40. Further, the option to move home results in lower consumption losses at all ages. This suggests that workers use the option to move home in order to shield consumption when wages fall on job separation.

When I examine only workers who move home on an EU transition, in that they live independently when employed and live at home when unemployed, I find further evidence of the move home option being used as a consumption smoothing mechanism. As shown in Table 8, an EU transition among older workers leads to a

Figure 9: Consumption Losses on Job Separation



**Note:** One way to examine the value of an insurance mechanism is to evaluate behaviour following a negative income shock (unemployment). In the first panel, high consumption losses when young are driven by a desire for asset accumulation. As workers age and become wealthier, they increasingly substitute assets to offset potential declines in consumption. When controlling for difference in income, it is clear that consumption losses on an unemployment shock are monotonically declining in age. However, there are still insurance benefits to co-residence at all ages. This can be seen through the vertical distance between consumption losses in the baseline model and in the counterfactual with no option to co-reside.

consumption loss, whereas for younger workers consumption still increases slightly. This is due to differences in future expectation of higher-wage periods that are declining in age. Among all ages, consumption while unemployed is higher in a model economy with the option to move home relative to a counterfactual where everyone lives independently. This indicates that the option to move home provides consumption insurance not just in the aggregate, but among those children who intend to move home on an EU transition.

Table 8: Consumption Change on EU Transition

| Age | Baseline Model (% $\Delta$ ) | No Co-Residence (% $\Delta$ ) |
|-----|------------------------------|-------------------------------|
| 31  | +14.9                        | -14.9                         |
| 32  | +8.9                         | -19.4                         |
| 33  | +5.4                         | -22.0                         |
| 34  | +3.4                         | -23.5                         |
| 35  | 0                            | -26.0                         |
| 36  | -3.6                         | -28.7                         |
| 37  | -5.5                         | -30.1                         |

**Note:** Consumption losses conditional on an EU transition in the baseline and no co-residence model environments. At some ages ( $< 35$ ), consumption rises as workers age regardless of employment status. This is due to positive expectations of future wages and differences in future opportunities to work based on time endowments. For older workers, a job separation is associated with consumption losses relative to the previous period. However, in an environment without the co-residence option all workers reduce consumption during an EU transition. At ages where consumption falls in both the baseline and counterfactual scenarios, losses are larger in the economy without the option to move home.

## 6 Conclusion

In this paper I extend our existing understanding of co-residence as non-market insurance to workers into middle age. By leveraging linked panel data from the Health and Retirement Study, I am able to account for a myriad of parent characteristics that are increasingly important in age and influence their child’s decision to move home. Previous analyses emphasize co-residence flows among young adults during post-secondary education or early in their careers. However, workers move home at similar rates following a job separation up to at least age 40, when workers are  $2.4\times$  as likely to move home relative to those who remain on-the-job. This finding is robust across differing levels of education, income, parental financial support, as well as parent income and assets. Crucially, I also control for health diagnoses and medical histories among parents, their functional limitations to work and living independently, and eldercare supplied by specific children.

To evaluate this insurance mechanism, I estimate a structural model of directed search and co-residence. In this environment, workers choose a submarket in which to search according to their own productivity and expected wages subject to an endogenous job-finding probability. Unemployment from an unsuccessful search is damaging primarily due to a fixed cost of housing that must be paid when agents live independently. Until age 40, workers can freely choose whether to pay this housing cost or avoid it by living at home at the cost of forgoing some utility from independence. This quantitative framework is able to reproduce the aggregate flows into co-residence among children in an Employed-to-Unemployed transition. By varying the

availability of the option to move home in the model economy, I find that when workers can co-reside they search for an additional eight weeks before finding a match. The option also provides significant welfare in this environment: in ex ante terms, being able to move home is equivalent to a 71% increase in the public unemployment insurance benefit. Consistent with the empirical results, older workers also value the insurance option highly; those aged 36-40 treat the option to move home as equivalent to a 79% increase in the UI benefit.

My findings have several implications for policy, both in the context unemployment insurance and elsewhere. On unemployment insurance, subsidies for shared housing may provide an alternative lever for policy makers as long as those subsidies can be structured more efficiently than direct cash transfers. As I have shown, co-residence between parents and children has significant implications for job search and match formation. Encouraging co-residence through subsidies, tax credits, or other public mechanisms will alter the job-finding environment. Further work is needed to evaluate the quality of matches emerging from search while co-residing to understand the tradeoffs associated with this mechanism.

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## A Appendix

### A.1 HRS: Pr(Move Home) Probit Estimation

Table 9 reports the full probit introduced in section 2.3. Individual observations correspond to pairs of parents and children. Standard errors are clustered at the pair-level and the McFadden's  $R^2$  is 0.1307. A joint Wald test of significance on all coefficients rejects the null hypothesis with extreme confidence. The formal specification is:

$$Pr(MoveHome) = \alpha_0 + \alpha \textit{Year} + \beta \textit{X}^c + \gamma \textit{X}^p + \delta \textit{X}^h + \eta \textit{X}^e + \epsilon$$

The vectors of characteristics are as follows:

- Child characteristics:  $\textit{X}^c = \{\text{Income, Education, Age, Gender, Employment Transition, Marital Status, Parental Status}\}$
- Parent characteristics:  $\textit{X}^p = \{\text{Income, Assets, Primary Residence Value, Education, Employment}\}$
- Parent health characteristics:  $\textit{X}^h = \{\text{High Blood Pressure, Diabetes, Lung Disease, Heart Disease, Ever Had Stroke, Ever Had Cancer}\}$
- Eldercare characteristics:  $\textit{X}^e = \{\text{Child Helps with ADLs, Child Helps with IADLs, Child Intends to Help in Future}\}$

Table 9: Pr(Move Home) Probit: Full Estimation

| Independent Variable | AME         | p-value |
|----------------------|-------------|---------|
| <b>Year</b>          |             |         |
| 2000                 | <i>Base</i> |         |
| 2002                 | 0.00246     | 0.182   |
| 2004                 | 0.00483     | 0.023   |
| 2006                 | 0.00024     | 0.898   |
| 2008                 | 0.00738     | 0.001   |
| 2010                 | 0.00369     | 0.076   |
| 2012                 | 0.00466     | 0.016   |

*Continued on next page*



Table 9: Average Marginal Effects on Pr(Move Home) (continued)

| Independent Variable               | AME                    | p-value |
|------------------------------------|------------------------|---------|
| 2014                               | 0.00504                | 0.020   |
| 2016                               | 0.00782                | 0.000   |
| 2018                               | 0.01095                | 0.000   |
| <b>Child Income</b>                |                        |         |
| <10K                               | <i>Base</i>            |         |
| 10–35K                             | −0.01155               | 0.000   |
| 35–70K                             | −0.02003               | 0.000   |
| 70–100K                            | −0.02035               | 0.000   |
| 100K+                              | −0.02411               | 0.000   |
| <b>Child Education</b>             |                        |         |
| Less Than High School              | <i>Base</i>            |         |
| High School Graduate               | −0.00433               | 0.193   |
| Partial College Completion         | 0.00203                | 0.563   |
| College and Above                  | −0.00141               | 0.669   |
| <b>Child Age</b>                   |                        |         |
| 20–25                              | <i>Base</i>            |         |
| 26–30                              | −0.00629               | 0.001   |
| 31–35                              | −0.00837               | 0.000   |
| 36–40                              | −0.01244               | 0.000   |
| <b>Child Employment Transition</b> |                        |         |
| E–E                                | <i>Base</i>            |         |
| E–U                                | 0.01588                | 0.000   |
| U–E                                | 0.00353                | 0.040   |
| U–U                                | 0.00331                | 0.079   |
| Child Gender (Female)              | −0.00338               | 0.000   |
| Child Marital Status               | −0.02618               | 0.000   |
| Child Parental Status              | 0.00034                | 0.790   |
| Child Helps w/ ADLs                | 0.01627                | 0.007   |
| Child Helps w/ IADLs               | 0.01269                | 0.007   |
| Child Intends to Help in Future    | 0.01125                | 0.000   |
| Parent Income                      | $6.22 \times 10^{-10}$ | 0.802   |
| Parent Assets                      | $-1.84 \times 10^{-9}$ | 0.021   |
| Parent Home Value                  | $3.86 \times 10^{-9}$  | 0.002   |
| <b>Parent Education</b>            |                        |         |
| Less Than High School              | <i>Base</i>            |         |
| GED                                | −0.00300               | 0.263   |
| High School Graduate               | 0.00412                | 0.028   |
| Partial College Completion         | 0.00276                | 0.135   |
| College and Above                  | 0.00397                | 0.037   |

*Continued on next page*

Table 9: Average Marginal Effects on Pr(Move Home) (continued)

| Independent Variable           | AME      | p-value     |
|--------------------------------|----------|-------------|
| <b>Parent Employment</b>       |          |             |
| Not Working                    |          | <i>Base</i> |
| Part-time Work                 | 0.00160  | 0.457       |
| Full-time Work                 | 0.00318  | 0.008       |
| Parent has High Blood Pressure | −0.00091 | 0.435       |
| Parent has Diabetes            | −0.00007 | 0.960       |
| Parent has ever had Cancer     | −0.00107 | 0.464       |
| Parent has Lung Disease        | 0.00067  | 0.689       |
| Parent has Heart Disease       | 0.00069  | 0.583       |
| Parent has ever had a Stroke   | 0.00518  | 0.005       |

**Note:** Average marginal effects are expressed in relative deviations from the mean share of children who move home (0.007518). McFadden’s  $R^2$ : 0.1307; Wald  $\chi^2$ : 1236.79

## A.2 NLSY97: Search Duration and Co-Residence

Table 10 reports the full OLS specification first presented in section 2.5. I estimate a regression equation of the form:

$$\begin{aligned}
 SearchDuration = & \beta_0 + \beta_1 CoResidence + \beta_2 Year + \beta_3 Age + \beta_4 Education \\
 & + \beta_5 Gender + \beta_6 TransferfromParent + \epsilon
 \end{aligned}$$

Where **Year**, **Age**, and **Education** are vectors of indicator variables corresponding to  $\beta_2$ ,  $\beta_3$ , and  $\beta_4$  vectors of coefficients for those indicators, respectively.

Age and year fixed-effects are generally not statistically significant with one exception: 2009. This is not entirely surprising, as workers at this time were experiencing the effects of the Great Financial Crisis, and as during any recession the most severe employment effects are disproportionately borne by low-productivity and low-tenure workers. These estimates are relatively imprecise, but the marginal effect of living at home on search duration are significant and economically meaningful. Additional measures of overall model fit such as the coefficient of determination ( $R^2$ ) and a joint test of coefficient significance ( $F$ -test) indicate the model is well estimated.

Table 10: OLS Estimation of Co-Residence and Search Duration: Full Results

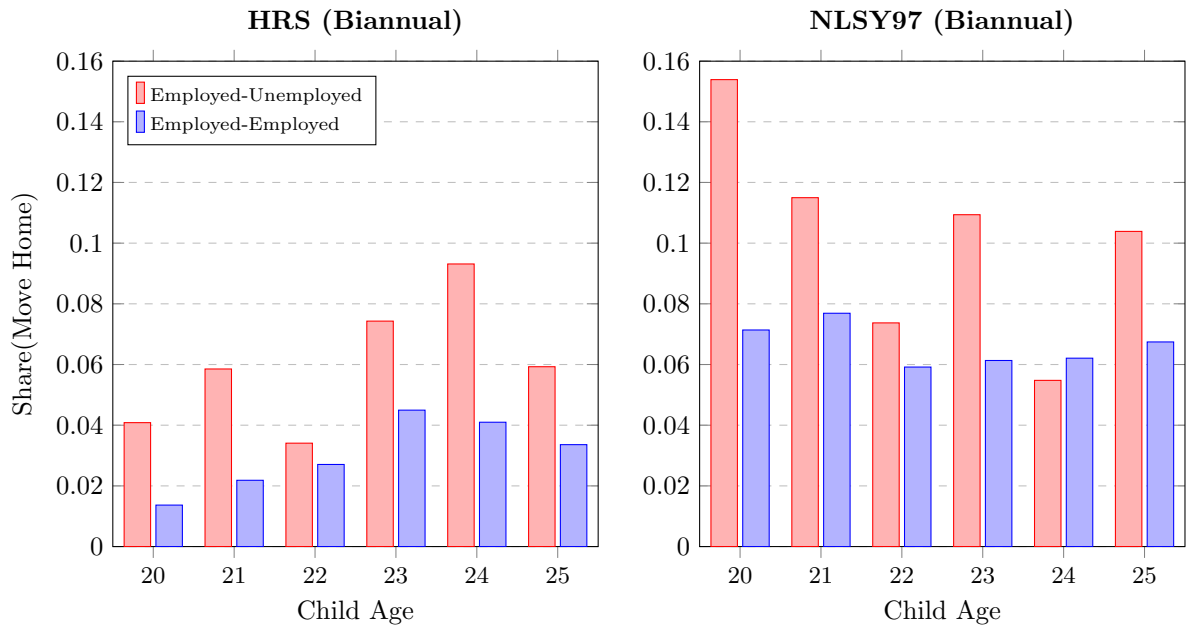
| Independent Variable           | Coefficient | p-value |
|--------------------------------|-------------|---------|
| Co-Residence                   | 1.389       | 0.034   |
| <b>Year</b>                    |             |         |
| 1999                           | <i>Base</i> |         |
| 2000                           | 0.840       | 0.657   |
| 2001                           | 0.875       | 0.617   |
| 2002                           | 1.195       | 0.480   |
| 2004                           | 1.652       | 0.376   |
| 2005                           | 0.671       | 0.741   |
| 2006                           | −0.448      | 0.827   |
| 2007                           | −0.230      | 0.914   |
| 2008                           | 2.528       | 0.245   |
| 2009                           | 7.489       | 0.001   |
| <b>Age</b>                     |             |         |
| 18                             | <i>Base</i> |         |
| 19                             | 0.078       | 0.9840  |
| 20                             | 0.561       | 0.883   |
| 21                             | 1.584       | 0.677   |
| 22                             | 3.304       | 0.399   |
| 23                             | 3.203       | 0.425   |
| 24                             | 3.181       | 0.419   |
| 25                             | 3.294       | 0.409   |
| 26                             | 2.598       | 0.518   |
| 27                             | 4.748       | 0.247   |
| 28                             | 2.683       | 0.519   |
| 29                             | 3.273       | 0.470   |
| <b>Education</b>               |             |         |
| Less Than High School          | <i>Base</i> |         |
| HS Graduate                    | −1.990      | 0.004   |
| Some College                   | −1.548      | 0.446   |
| Post-Secondary and Above       | −2.431      | 0.055   |
| Gender (Female)                | −0.970      | 0.055   |
| Financial Transfer from Parent | −1.114      | 0.112   |
| Constant                       | 8.030       | 0.054   |

**Note:**  $R^2$ : 0.0766,  $F$ -statistic: 4.43

### A.3 Robustness: HRS versus Biannual NLSY97

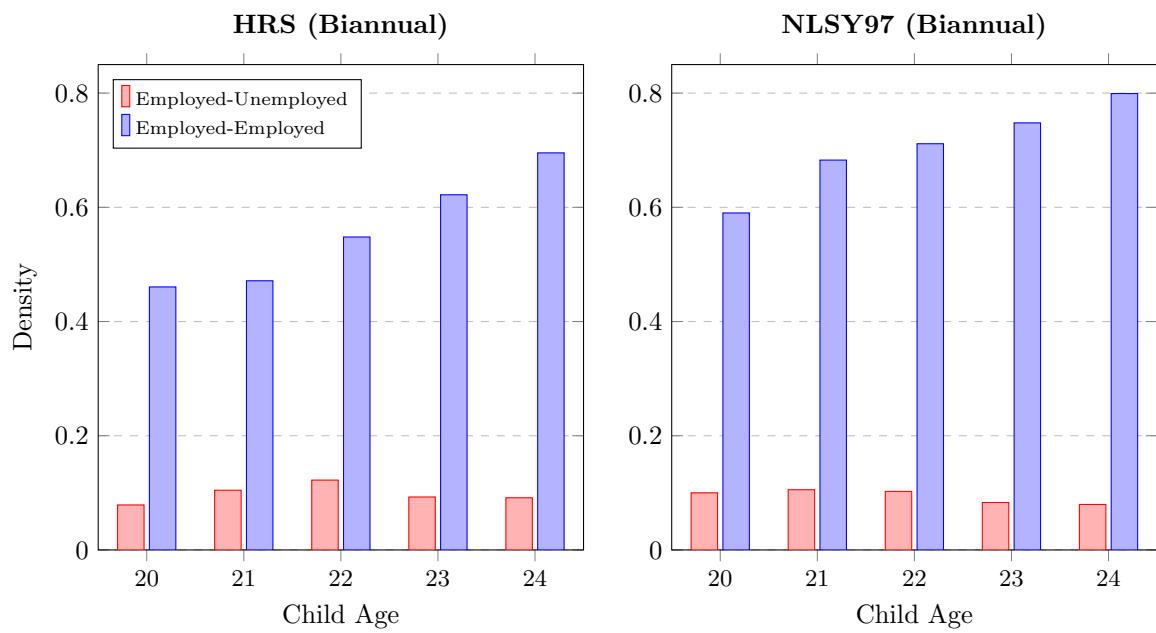
One limitation to comparing moments from the HRS and NLSY97 is the differences in frequency between the two surveys. This is relatively straightforward to correct simply by transforming the NLSY into a biannual dataset. This process is akin dropping approximately half of the observations from the NLSY97 and is thus very costly from the perspective of statistical precision. Nonetheless, this transformation can be valuable to understand the role of survey frequency in differences in moments between the two sources. Figure 10 shows the share of workers who move home conditional on EU and EE transitions. In section 2.6, I pointed out a difference in levels between the two datasets. Adjusting the NLSY97 to a biannual frequency exacerbates these differences in levels, resulting in move home shares in the NLSY97 that are significantly larger than when treating the data annually. Figure 11 report employment flows under a similar adjustment.

Figure 10: Comparison of Move Home Trends Among the HRS and NLSY97



**Note:** EU workers are more likely to move home relative to their EE counterparts. Adjusting the NLSY97 to mimic the biannual nature of the HRS does not shrink the differences in levels between the two datasets, but instead increases them.

Figure 11: Comparison of Employment Transitions Between HRS and NLSY97 (2004)



**Note:** Biannually transforming the NLSY97 reduces the difference in levels slightly, but comparability issues persist. As discussed in section 2.6, these issues likely arise from differences in sample design.